Speaker 2 - your script

Slides 4-6: DQN, the paper's idea, attention. About 1 min 55 s. Speak slowly, pause at full stops, and look up at the TA between sentences - you don't have to read every word exactly.

Slide 4

[Take the clicker. Click to slide 4.]

Thanks. Now, there are far too many situations to write all the scores in a table. So we use a neural network to estimate them. This is called DQN, a Deep Q-Network. To keep the learning stable, it uses three simple tricks. First, it remembers past moments and studies them in random order, a bit like revising old notes. Second, it checks its answers against a slowly updated copy of itself, so the target doesn't keep moving. And third, at the start it tries random moves, just to explore.

Slide 5

[Click to slide 5. Point at the table row by row: FCN, CNN, Attention.]

Now here's the real question the paper asks: how should the car look at the other cars? It compares three ideas. FCN reads all the cars as one long list of numbers, but the list gets shuffled, so it can't tell which car is which. CNN looks at a small, blurry picture of the road from above. And attention, the paper's idea, looks at every car and asks: how important is this one for me? To keep it fair, all three networks are about the same size.

Slide 6

[Click to slide 6. Point at the 72 % row, then at the picture with the lines.]

Attention is actually very close to how we drive. Think about it: when you cross a road, you don't look at every car equally. You focus on the one coming fast towards you. Our car does the same. It gives each car a danger score, and softmax turns those scores into percentages. Here, the fast car from the left gets seventy-two percent of the attention, and the rest get very little. And the nice part is, we can actually see what it's looking at, like the lines in this picture. Now my teammate will show you how well it works.

[Hand over to Speaker 3.]